WIND FARM STUDY REPORT

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Program: European Masters in Renewable Energy

Course: Wind Energy

Code: 061

1. Analysis of Wind Data Measurements.

Wind Measurements comprising of wind speed and wind direction from site M8 were collected and then the annual average wind speed and the standard deviation were calculated, and it was approximately around 6.8m/sec and 3.57 respectively. The following wind speeds of the site were then grouped starting from 0m/sec to 38m/sec and the frequency of each wind speed that appeared in the site was calculated using excel formula (countifs). The 16 possible directions of the wind speed were noted down. After calculating the possible number of outcomes of each wind speeds a radar chart was plotted using the information of the ranges of wind speed that occurred in the site and the cumulative frequency of wind speed in each direction. This radar chart represents the wind rose in 16 directions. For plotting the Weibull distribution, the scale factor and shape factor calculation is important, generally two methods are used for calculation shape and scale factor, semi-empirical formula of Bowden and the least square method. The least square method is more effective, and it gave a more precise result (k=1.95 and c=7.351). The Bowden formula calculated the shape factor and scale factor as (k=1.9 and c=7.45). Now as the shape factor and scale factor has been calculated, using these values the average wind speed of the site can be calculated and it was approximately around 7.1m/sec. Both the average wind speed is almost equal with negligible differences. Finally, the average annual wind power density was calculated, and it was approximately equal to 213.84 kW.

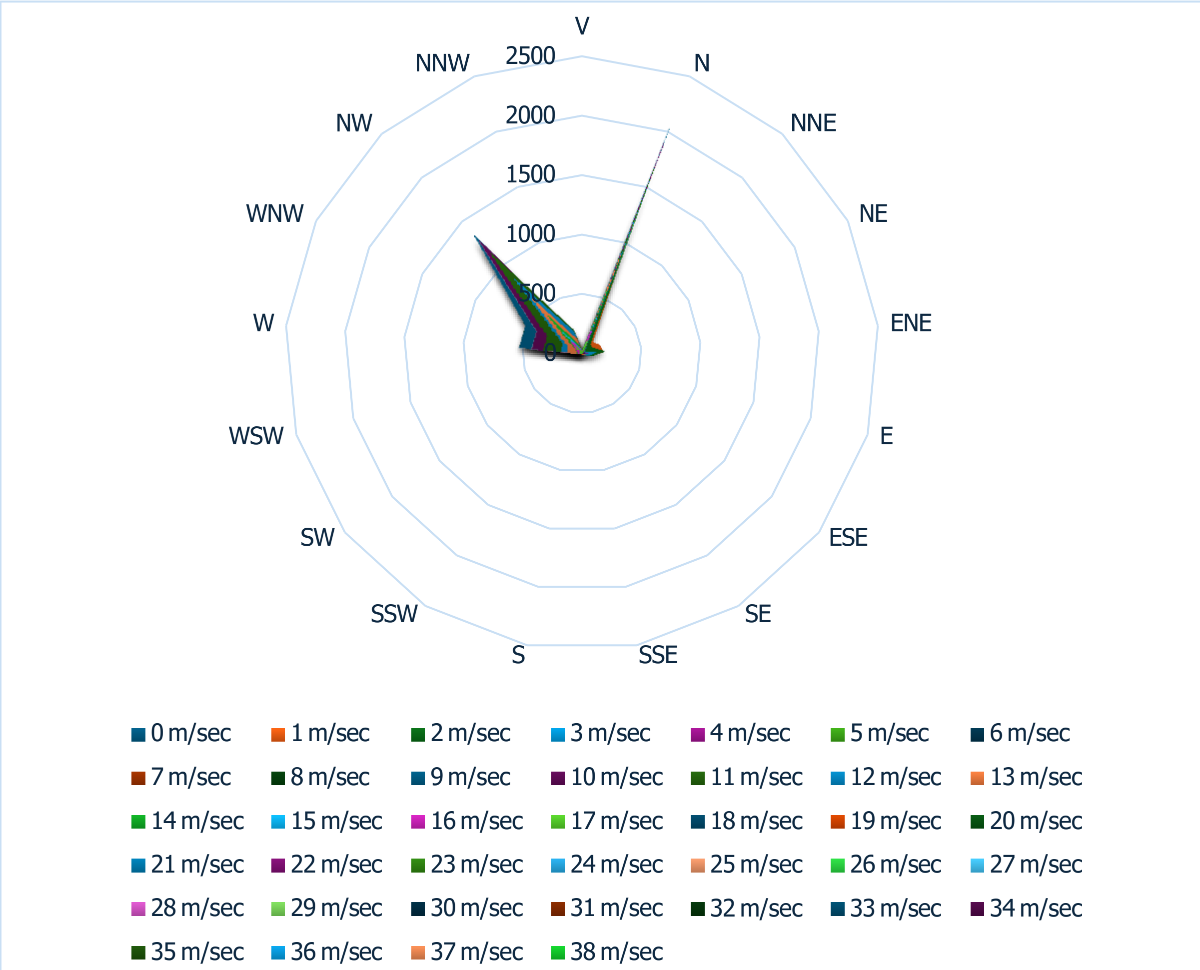
Average wind Speed	Standard Deviation
6.8	3.57

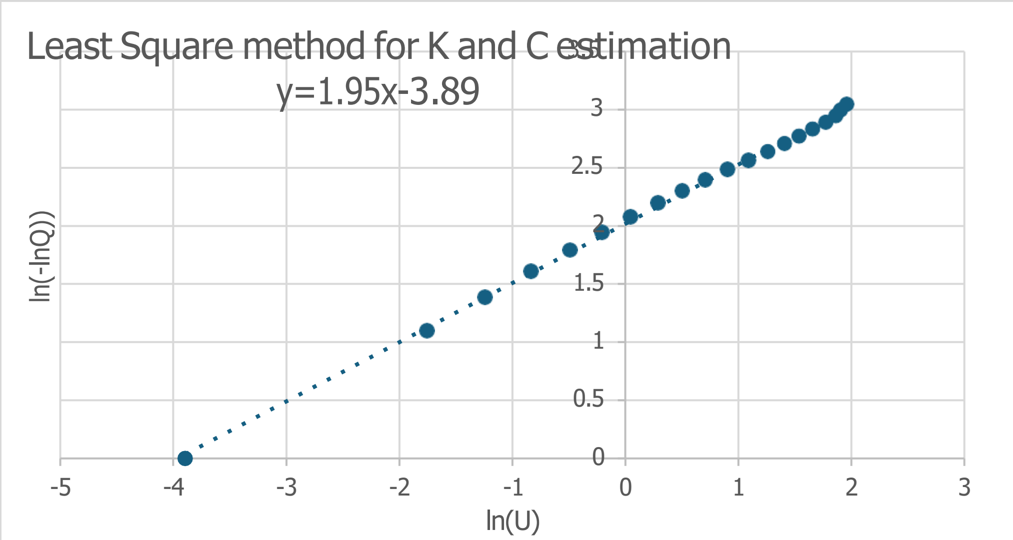
V	least squa	bowden	V	Power density	
0	0	0	0	0	
1	0.03907	0.04093	1	7.55356	
2	0.07119	0.07193	2	18.393	
3	0.09514	0.09418	3	21.2047	213.843
4	0.10968	0.10722	4	21.2047	
5	0.11479	0.11147	5	22.1939	
6	0.1116	0.10818	6	21.578	
7	0.10203	0.09918	7	19.73	
8	0.08839	0.08654	8	17.0926	
9	0.07289	0.0722	9	14.0983	
10	0.05743	0.05779	10	11.1087	
11	0.04333	0.04449	11	8.38205	
12	0.03136	0.03301	12	6.06769	
13	0.02181	0.02363	13	4.21982	
14	0.01458	0.01634	14	2.82258	
15	0.00939	0.01093	15	1.81749	
16	0.00582	0.00707	16	1.12745	
17	0.00348	0.00443	17	0.67419	
18	0.00201	0.00269	18	0.38883	
19	0.00112	0.00159	19	0.21639	
20	0.0006	0.00091	20	0.11624	
21	0.00031	0.0005	21	0.0603	
22	0.00016	0.00027	22	0.03021	
23	7.5E-05	0.00014	23	13.7627	

Table for the Least Square and Bowden weibull distribution and power density

11.25	33.75	56.25	78.75	101.25	123.75	146.25	168.75	191.25	213.75	236.25	258.75	281.25	303.75	326.25	348.75	
348.75	11.25	33.75	56.25	78.75	101.25	123.75	146.25	168.75	191.25	213.75	236.25	258.75	281.25	303.75	326.25	
V	N	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW
0	171	25	35	28	0	0	0	0	0	0	0	0	7	7	48	28
1	1022	126	162	175	33	0	0	0	0	0	0	0	63	63	303	160
2	1553	97	97	183	76	0	0	0	0	0	0	0	136	136	759	205
3	1595	28	25	116	61	0	0	0	0	0	0	0	308	308	973	84
4	1462	16	13	53	18	0	0	0	0	0	0	0	382	382	903	77
5	1476	2	2	25	18	0	0	0	0	0	0	0	436	436	886	107
6	1553	0	1	22	21	0	0	0	0	0	0	0	461	461	903	145
7	1746	0	0	20	17	0	0	0	0	0	0	0	500	500	1091	118
8	1976	0	0	15	7	0	0	0	0	0	0	0	525	525	1298	131
9	2039	1	0	18	5	0	0	0	0	0	0	0	526	526	1341	148
10	1905	0	0	16	2	0	0	0	0	0	0	0	424	424	1287	176
11	1691	0	0	16	0	0	0	0	0	0	0	0	325	325	1138	212
12	1233	0	0	7	0	0	0	0	0	0	0	0	187	187	844	195
13	880	0	0	1	1	0	0	0	0	0	0	0	128	128	624	126
14	567	0	1	2	0	0	0	0	0	0	0	0	54	54	431	79
15	319	0	0	1	0	0	0	0	0	0	0	0	27	27	244	47
16	208	0	1	4	0	0	0	0	0	0	0	0	38	38	137	28
17	117	0	2	1	0	0	0	0	0	0	0	0	13	13	79	22
18	60	0	0	1	0	0	0	0	0	0	0	0	21	21	31	7
19	17	0	1	0	0	0	0	0	0	0	0	0	5	5	10	1
20	11	0	1	3	0	0	0	0	0	0	0	0	4	4	3	0
21	3	0	3	0	0	0	0	0	0	0	0	0	0	0	0	0
22	1	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0
23	6	0	5	1	0	0	0	0	0	0	0	0	0	0	0	0
24	7	1	4	2	0	0	0	0	0	0	0	0	0	0	0	0
25	2	0	2	0	0	0	0	0	0	0	0	0	0	0	0	0
26	3	0	0	3	0	0	0	0	0	0	0	0	0	0	0	0
27	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
28	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
29	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
30	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
31	1	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0
32	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
33	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
34	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
35	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
36	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
37	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
38	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Table for the wind rose in 16 directions



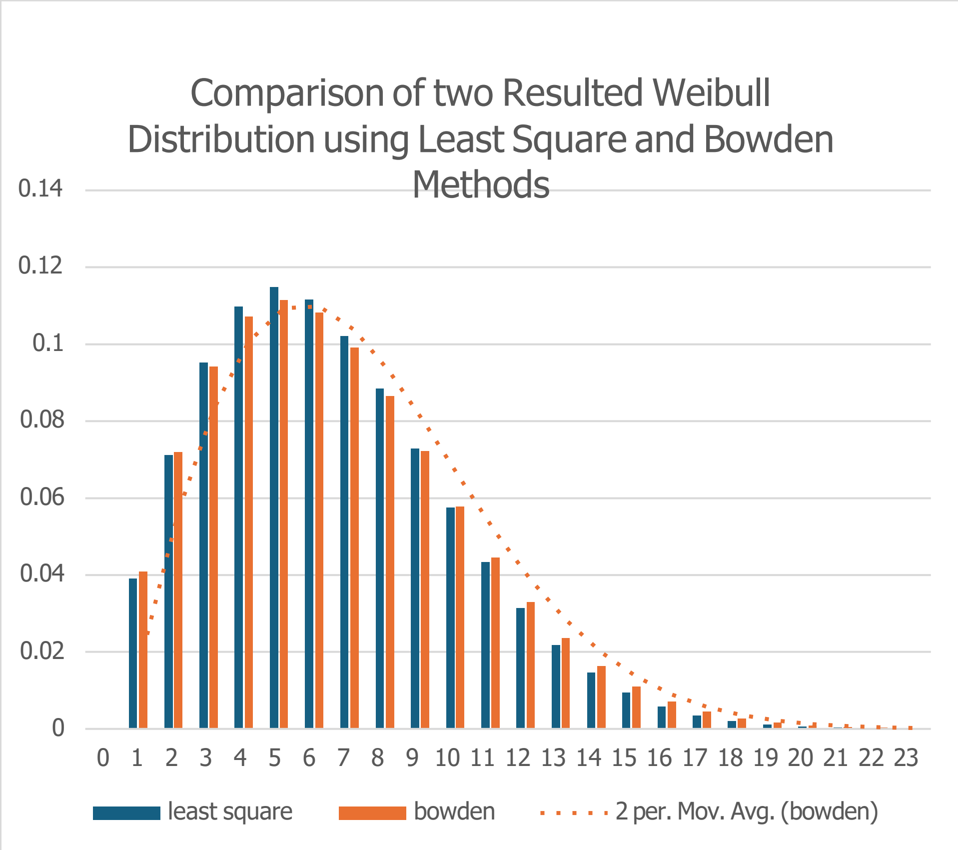


	Shape factor (k)	Scale factor (c)
Bowden	1.9	7.45
Least Square	1.95	7.35

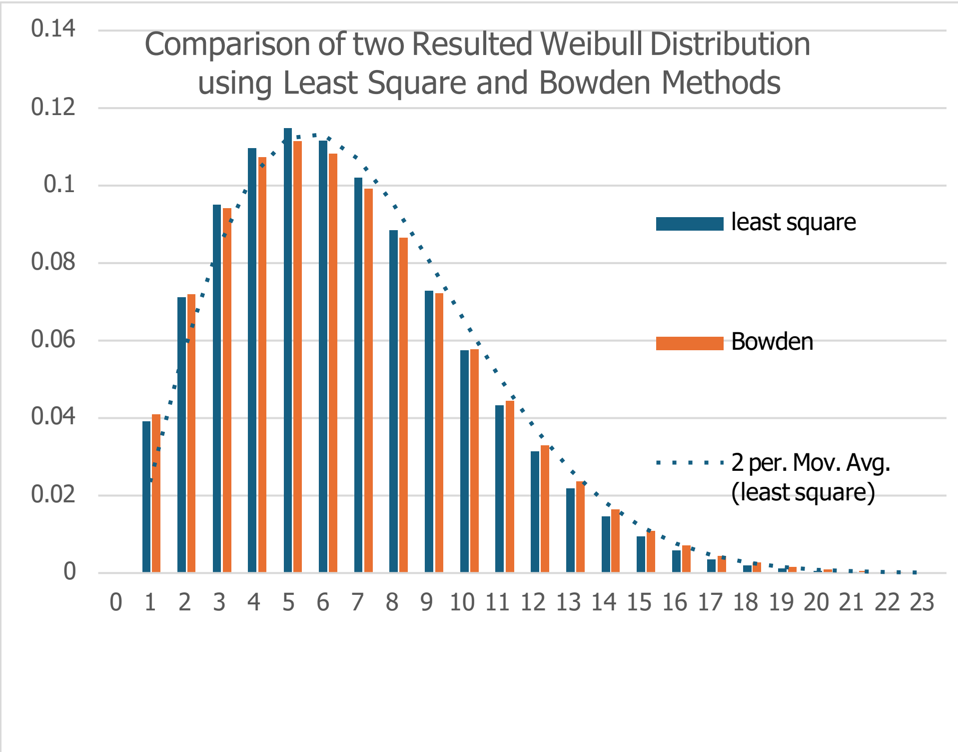
Average speed of wind in the site using k and c of Weibull Distribution	Average annual wind power density (Kw)
7.1	213.84

			Y	X		
0	51543	0.99994				
1	50507	0.97984	-3.8941	0		
2	47447	0.92048	-2.4906	0.69315		
3	43362	0.84123	-1.7551	1.09861		k
4	38594	0.74873	-1.24	1.38629		
5	33357	0.64713	-0.8319	1.60944		
6	27907	0.5414	-0.4884	1.79176		
7	22819	0.44269	-0.2047	1.94591		
8	18016	0.34951	0.04995	2.07944		
9	13571	0.26328	0.28859	2.19722		
10	9811	0.19033	0.5062	2.30259		
11	6756	0.13107	0.70904	2.3979		
12	4318	0.08377	0.90813	2.48491		
13	2628	0.05098	1.09066	2.56495		
14	1495	0.029	1.26423	2.63906		
15	857	0.01663	1.4102	2.70805		
16	487	0.00945	1.53944	2.77259		
17	271	0.00526	1.65787	2.83321		
18	143	0.00277	1.77281	2.89037		
19	82	0.00159	1.86307	2.94444		
20	62	0.0012	1.90555	2.99573		
21	43	0.00083	1.95855	3.04452		
22	33	0.00064	-7.3534	3.09104		

Table for the Y and X calculation in least square method used for calculating K & C



Comparison of two Resulted Weibull Distribution using Least Square and Bowden Methods
0.14
0.12
0.1
0.08
0.06
0.04
0.02
0
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23
least square
bowden
2 per. Mov. Avg. (bowden)



Comparison of two Resulted Weibull Distribution using Least Square and Bowden Methods
0.14
0.12
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0.04
0.02
0
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23
least square
Bowden
2 per. Mov. Avg. (least square)

2. Wind Turbine Power Curve Assessment.

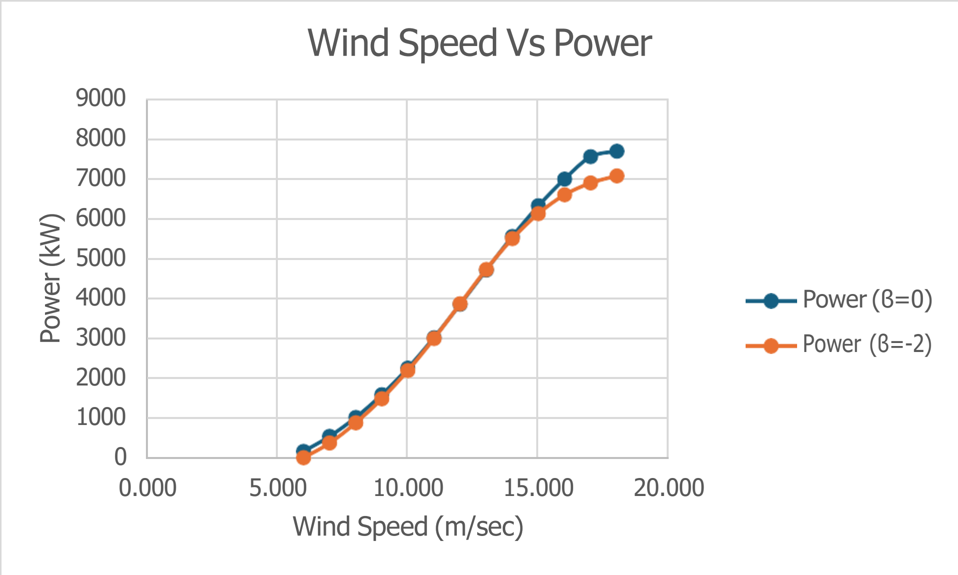
All the measurements of chord radius twist thickness of blade are given in the excel sheet. A radius was assumed (for example 50m), and all the values of the radius and the chord given in the excel were multiplied with the assumed radius. A notepad file comprising the geometric data of the blade was created. After the geometric section was completed, interpolation was done where some extra thicknesses were introduced, and calculations were done for the respective chord and radius based on those thicknesses. Another notepad file was created where the profile of the blade was stored (radius along with the thickness). Finally, the aerodynamic file was created where the wind speeds were adjusted, and the rotational speed was calculated, and other parameters were kept constant including the rotational speed. The other files (geometric and profile file) were called in the aerodynamic file and then the raft application was executed. The load file was created were the power, tip speed ratio and the thrust were calculated with variable wind speeds but fixed pitch angle and rotational speed. The wind speed and the power values were copied from the notepad to the excel sheet and the power coefficients were calculated. The maximum power coefficient was 0.47 with an optimum tip speed ratio of 7.27. Graphs of wind speed Vs turbine power and the tip speed ratio Vs power coefficients were created. After the graphs were created the maximum power, the designed power and real power were calculated based on the rated power of the wind turbine. The maximum and minimum wind speed (11.00 m/sec and 6.60 m/sec), the maximum and minimum rotational speeds (1.6 rad/sec, 0.96 rad/sec) were all calculated. The radius was then changed from 50m to 60m. Graphs based on wind speed and the maximum and real power were plotted. The scale factor was calculated as the square of the new radius divided by the previous rated radius of the turbine. After the scale factor is calculated then it is multiplied with the power, maximum power and real power of the turbine having 50m radius and then new graphs are plotted showing the difference in the power and wind speed curve depending on radius and the rated power as a horizontal line parallel to the x axis. Graphs were plotted depicting the changes in the wind speed Vs real power and maximum power curves of wind turbines having 50m and 60m radius (change of power curves with radius). In order to study the effect of the pitch angle on the power curves the pitch angle was changed from 0 to -2 in the aerodynamic notepad file and the raft application was executed again to create the new power, tip speed ratio, thrust values. This new power values against different wind speed with a change in the pitch angle were noted down. Graphs were plotted and then power coefficient was calculated and the new graph of tip speed ratio Vs power coefficient with a pitch angle -2 was compared to the previous one with a pitch angle 0. The maximum power coefficient has decreased to a smaller extent with the decrease in pitch angle. The following graphs of maximum power, real power was again created with the new pitch angle. The Weibull values that were calculated previously were noted down and the power of the wind turbine of the given radius was multiplied with and the summation of this product gave the result of the average power. The average power multiplied with the number hours in a year (8760 hrs) gave us the annual energy production. The capacity factor was calculated as the average power divided by the rated power of the wind turbine. The investment cost was noted from an excel file and other expenses were added to it for obtaining the total capital cost. The operation and maintenance cost were assumed as 3% of the total investment cost. Finally, the R was calculated as 0.087. Following the formula of the levelized cost the LCoE was calculated for wind turbines with different radius and graphs were plotted to analyze the relationship between the levelized cost and the radius of the wind turbines. Graphs showing the relationship between the pitch angle and the wind speed and the relationship between the wind speed and the rotational speed.

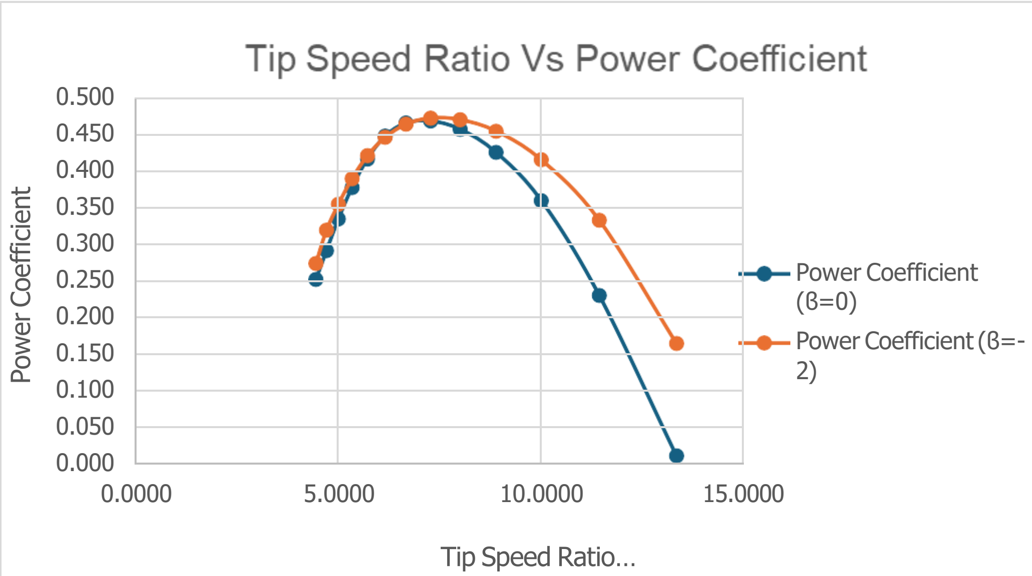
Wind Speed	Power (ϐ=0)	Wind Power
6.000	171	1038.555
7.000	550	1649.187
8.000	1023	2461.760
9.000	1594	3505.123
10.000	2261	4808.125
11.000	3025	6399.614
12.000	3856	8308.440
13.000	4714	10563.451
14.000	5558	13193.495
15.000	6326	16227.422
16.000	6995	19694.080
17.000	7555	23622.318
18.000	7696	28040.985

Power and Wind Speed table

Wind Speed	Power (ϐ=-2	Wind Power		Lamda	Power Coefficient (ϐ=0)
6.000	12	1038.5550		13.3333	0.012
7.000	381	1649.1869		11.4286	0.231
8.000	886	2461.7600		10.0000	0.360
9.000	1492	3505.1231		8.8889	0.426
10.000	2199	4808.1250		8.0000	0.457
11.000	2999	6399.6144		7.2727	0.469
12.000	3871	8308.4400		6.6667	0.466
13.000	4735	10563.4506		6.1539	0.448
14.000	5500	13193.4950		5.7143	0.417
15.000	6130	16227.4219		5.3333	0.378
16.000	6600	19694.0800		5.0000	0.335
17.000	6895	23622.3181		4.7059	0.292
18.000	7078	28040.9850		4.4444	0.252

Power, wind speed, tip speed ratio, and power coefficient table



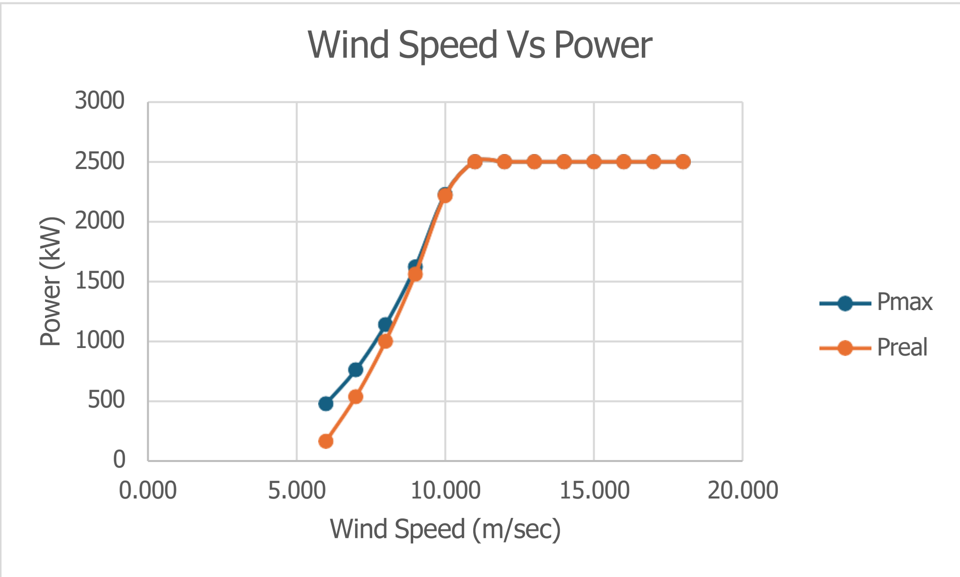


Lamda	Power Coeffi	Wind Speed	Power	Preal	Pmax	Pdesign	Omega
13.333	0.165	6.000	167.95004	167.95004	480.89721	472.16327	0.96
11.428	0.333	7.000	538.49535	538.49535	763.64697	720.93946	1.01808
10.000	0.416	8.000	1002.2831	1002.2831	1139.9045	1015.8967	1.16352
8.888	0.455	9.000	1561.2686	1561.2686	1623.0281	1338.53	1.30896
8.000	0.470	10.000	2215.2274	2215.2274	2226.376	1672.9114	1.6
7.272	0.473	11.000	2963.3064	2500	2500	2500	1.6
6.666	0.464	12.000	3777.3061	2500	2500	2500	1.6
6.153	0.446	13.000	4617.7959	2500	2500	2500	1.6
5.714	0.421	14.000	5444.5714	2500	2500	2500	1.6
5.333	0.390	15.000	6196.898	2500	2500	2500	1.6
5.000	0.355	16.000	6852.2449	2500	2500	2500	1.6
4.705	0.320	17.000	7400.8163	2500	2500	2500	1.6
4.444	0.274	18.000	7538.9388	2500	2500	2500	1.6

For pitch angle zero, and radius of the blade being 50m

Lamda	Power Coeffi	Wind Speed	Power	Preal	Pmax	Power	Pmax	Preal	Prated
13.333	0.165	6.000	167.95004	167.95004	480.89721	241.84806	692.49199	241.84806	2500
11.428	0.333	7.000	538.49535	538.49535	763.64697	775.4333	1099.6516	775.4333	2500
10.000	0.416	8.000	1002.2831	1002.2831	1139.9045	1443.2877	1641.4625	1443.2877	2500
8.888	0.455	9.000	1561.2686	1561.2686	1623.0281	2248.2267	2337.1605	2248.2267	2500
8.000	0.470	10.000	2215.2274	2215.2274	2226.376	3189.9275	2500	2500	2500
7.272	0.473	11.000	2963.3064	2500	2500	4267.1613	2500	2500	2500
6.666	0.464	12.000	3777.3061	2500	2500	5439.3208	2500	2500	2500
6.153	0.446	13.000	4617.7959	2500	2500	6649.6261	2500	2500	2500
5.714	0.421	14.000	5444.5714	2500	2500	7840.1829	2500	2500	2500
5.333	0.390	15.000	6196.898	2500	2500	8923.5331	2500	2500	2500
5.000	0.355	16.000	6852.2449	2500	2500	9867.2327	2500	2500	2500
4.705	0.320	17.000	7400.8163	2500	2500	10657.176	2500	2500	2500
4.444	0.274	18.000	7538.9388	2500	2500	10856.072	2500	2500	2500

For pitch angle zero and radius of blade being 50m and 60m



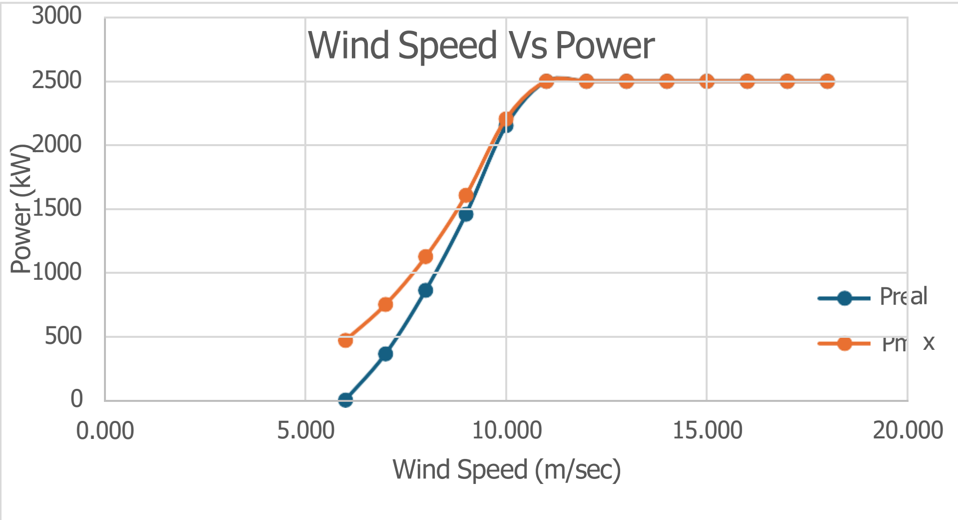
For pitch angle (β=0)

ind Speed	Lamda	Power Coeffi	Power	Preal	Pmax	Pdesign	Omega
6.000	13.333	0.0118621	12.068072	12.068072	476.69667	476.69667	0.96
7.000	11.429	0.2308857	373.00278	373.00278	756.97665	752.60924	1.01808
8.000	10.000	0.3600964	868.37973	868.37973	1129.9477	1080.9328	1.16352
9.000	8.889	0.4256627	1461.551	1461.551	1608.8512	1431.2943	1.30896
10.000	8.000	0.4573003	2153.8843	2153.8843	2206.929	1779.1841	1.6
11.000	7.273	0.4685624	2937.4225	2500	2500	2500	1.6
12.000	6.667	0.465859	3791.5707	2500	2500	2500	1.6
13.000	6.154	0.4482371	4638.2993	2500	2500	2500	1.6
14.000	5.714	0.4168507	5387.478	2500	2500	2500	1.6
15.000	5.333	0.3777461	6004.7462	2500	2500	2500	1.6
16.000	5.000	0.3351366	6465.5095	2500	2500	2500	1.6
17.000	4.706	0.2919033	6754.7092	2500	2500	2500	1.6
18.000	4.44444	0.2524061	6933.2726	2500	2500	2500	1.6

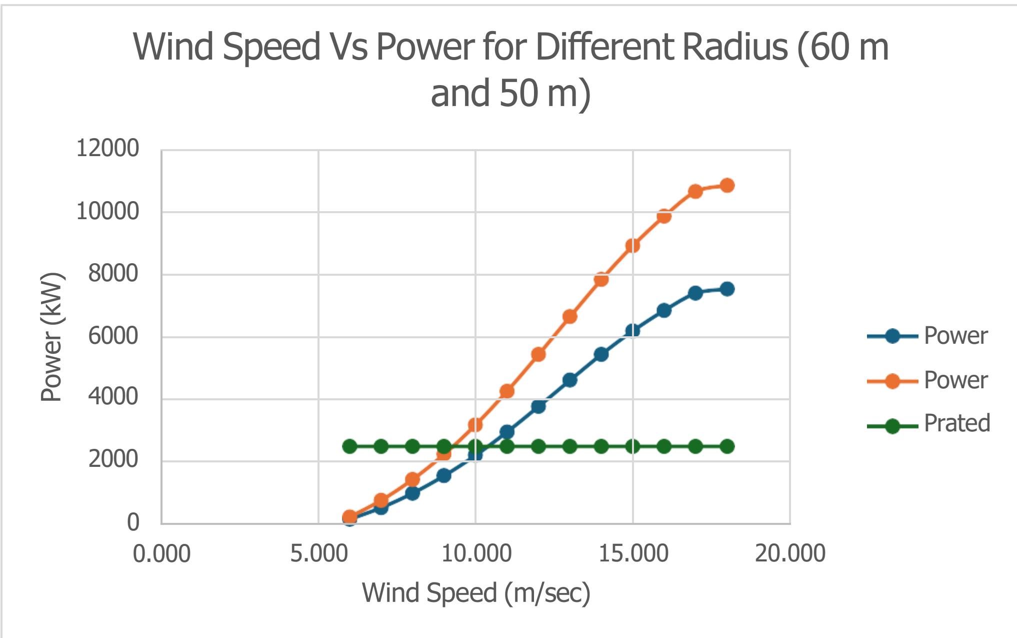
For pitch angle -2 and radius of blade being 50m

Wind Speed	Lamda	Power Coeffi	Power	Preal	Pmax		Power	Preal	Pmax	Prated
6.000	13.333	0.0118621	12.068072	12.068072	476.69667		17.378023	17.378023	686.4432	2500
7.000	11.429	0.2308857	373.00278	373.00278	756.97665		537.124	537.124	1090.0464	2500
8.000	10.000	0.3600964	868.37973	868.37973	1129.9477		1250.4668	1250.4668	1627.1246	2500
9.000	8.889	0.4256627	1461.551	1461.551	1608.8512		2104.6335	2104.6335	2316.7458	2500
10.000	8.000	0.4573003	2153.8843	2153.8843	2206.929		3101.5934	2500	2500	2500
11.000	7.273	0.4685624	2937.4225	2500	2500		4229.8884	2500	2500	2500
12.000	6.667	0.465859	3791.5707	2500	2500		5459.8619	2500	2500	2500
13.000	6.154	0.4482371	4638.2993	2500	2500		6679.151	2500	2500	2500
14.000	5.714	0.4168507	5387.478	2500	2500		7757.9683	2500	2500	2500
15.000	5.333	0.3777461	6004.7462	2500	2500		8646.8345	2500	2500	2500
16.000	5.000	0.3351366	6465.5095	2500	2500		9310.3336	2500	2500	2500
17.000	4.706	0.2919033	6754.7092	2500	2500		9726.7812	2500	2500	2500
18.000	4.44444	0.2524061	6933.2726	2500	2500		9983.9125	2500	2500	2500

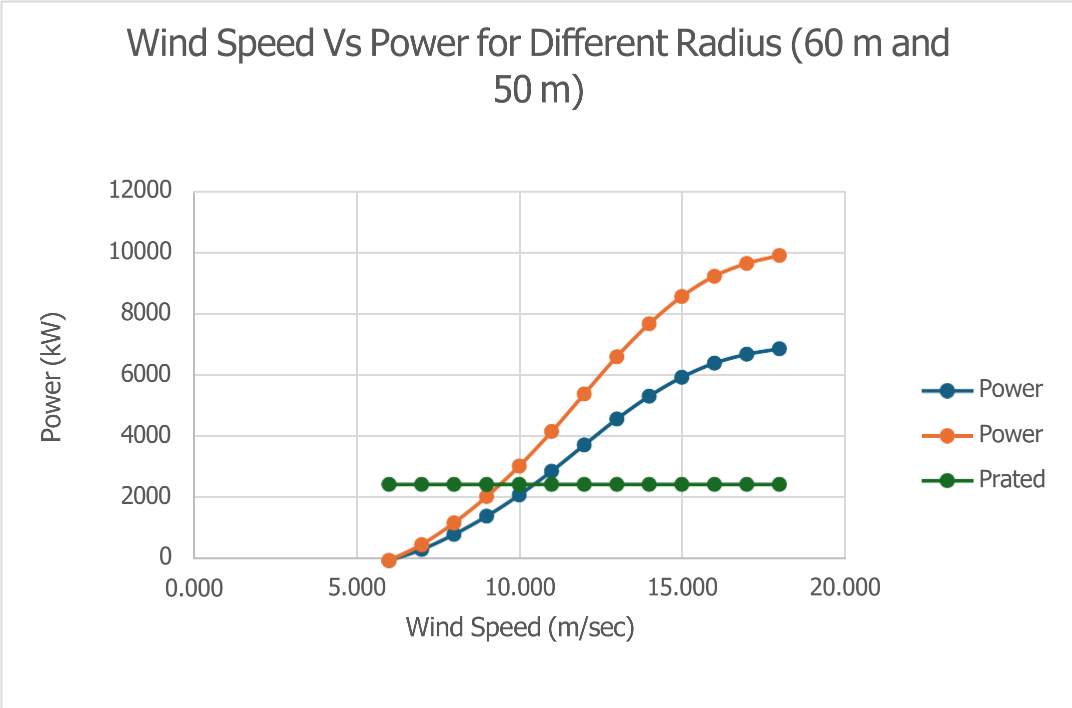
For pitch angle -2 and radius of blade being 50m and 60m



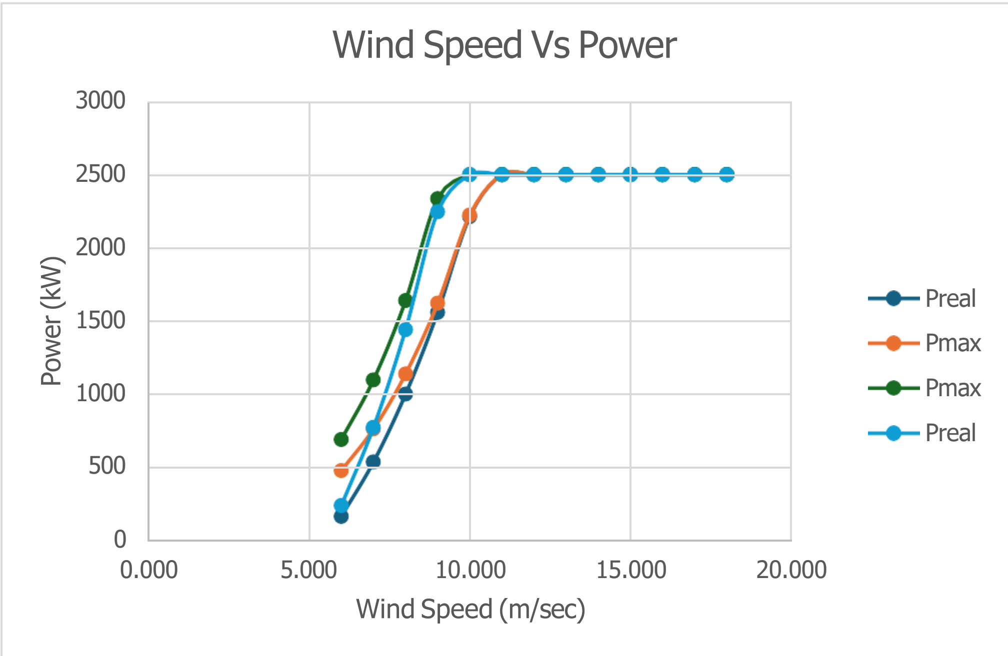
For pitch angle (β=-2)



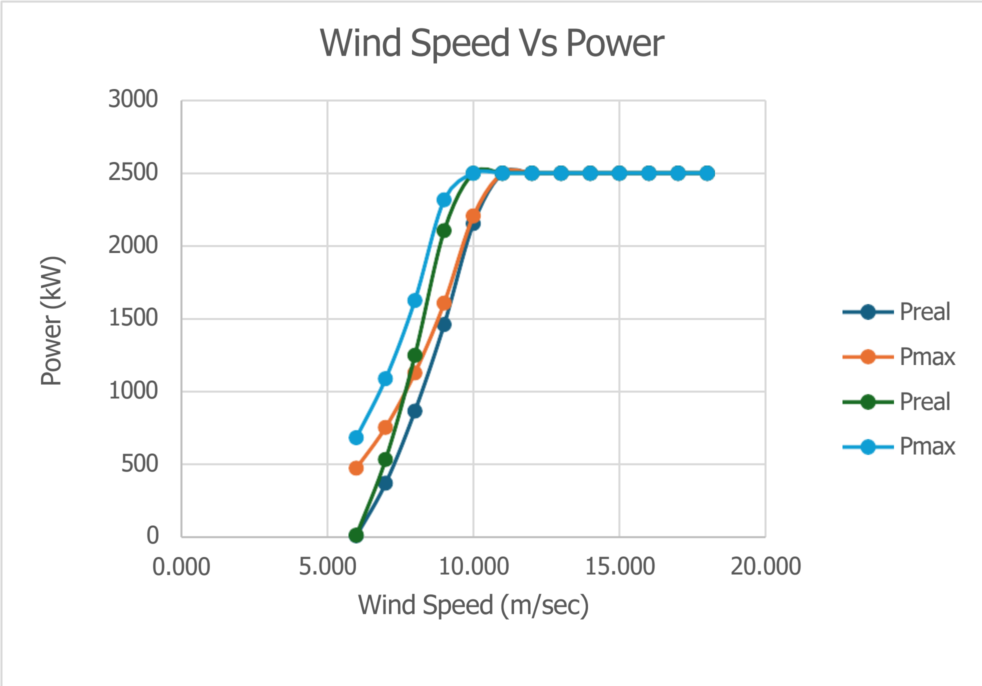
For pitch angle (β=0)



For pitch angle (β=-2)



For pitch angle (β=0)

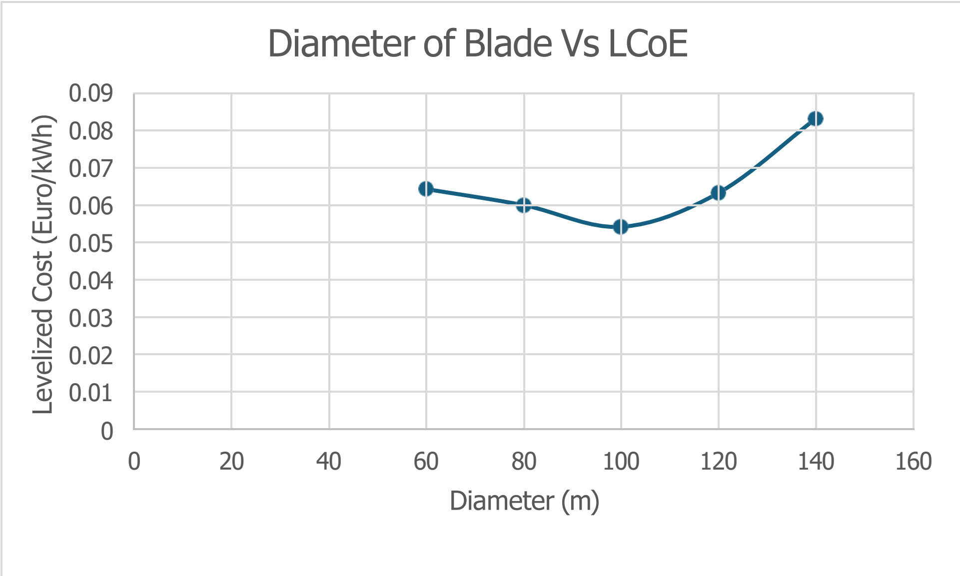


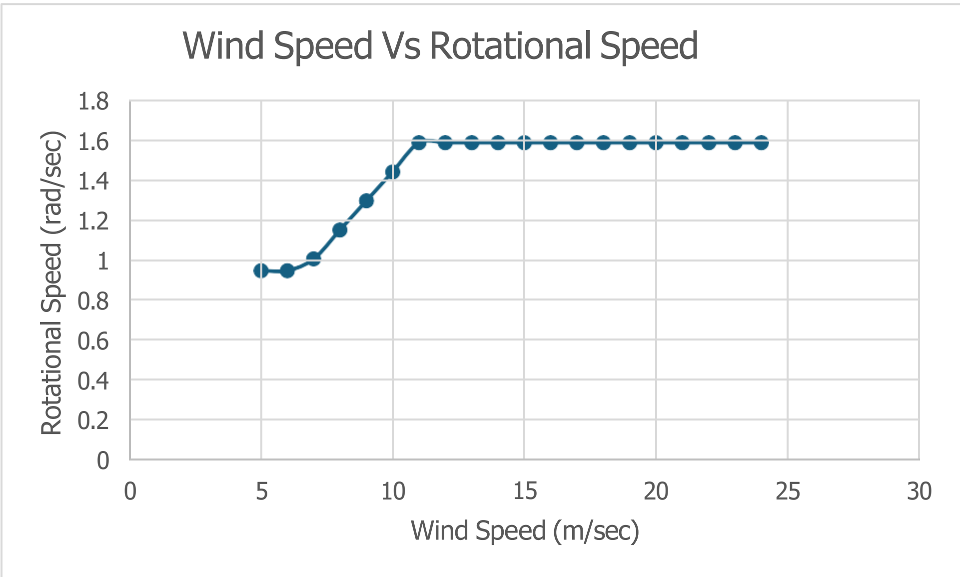
For pitch angle (β=-2)

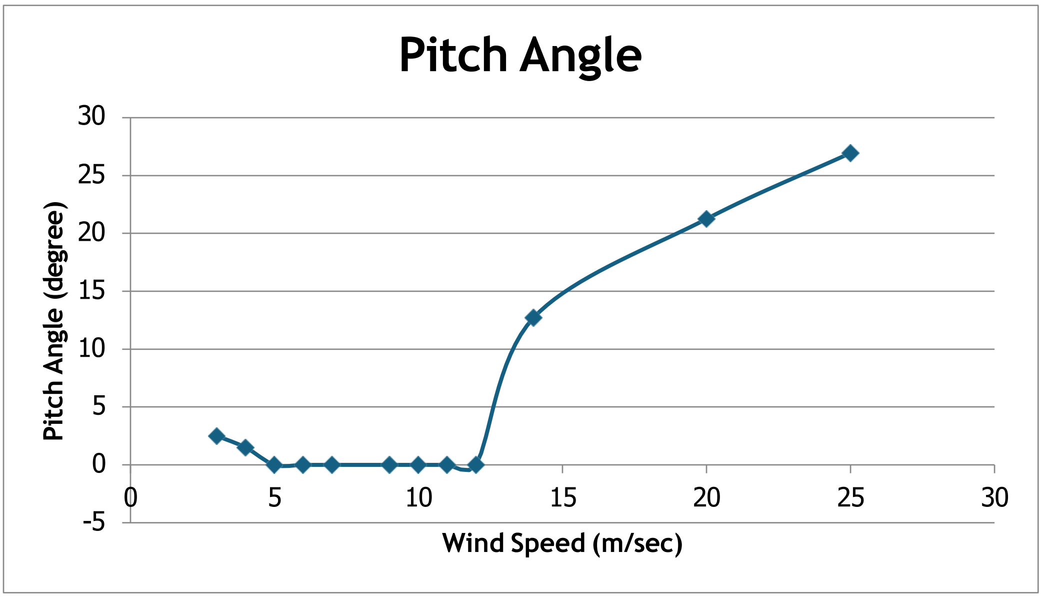
Capital cost	Taken from cost model
Annual Energy Production	Average power * 8760
Other costs	550*rated power(2.5)
Total Investment Cost	Capital costs + Other costs
Operational Cost	3% of Investments
R	0.087
LCoE	(R*Investment Cost + Operational Cost)/Annual Energy Production

Table for the LCoE calculations

D	LCoE
40	0.0649885
60	0.0649885
80	0.06068824
100	0.05490174
120	0.06395489
140	0.08382073

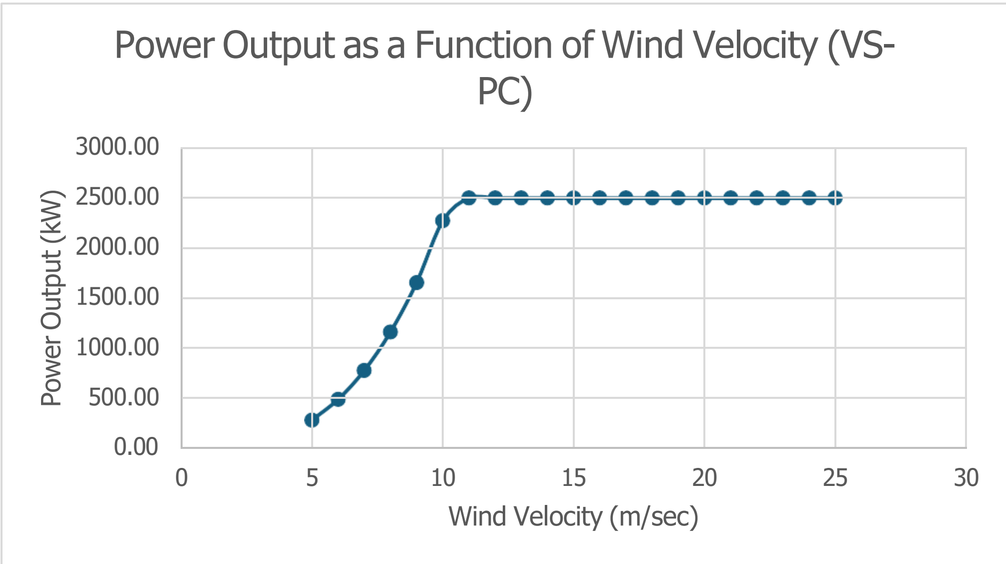

Diameter of Blade Vs LCoE
Levelized Cost (Euro/kWh)
0.09
0.08
0.07
0.06
0.05
0.04
0.03
0.02
0.01
0
0
20
40
60
80
100
120
140
160
Diameter (m)

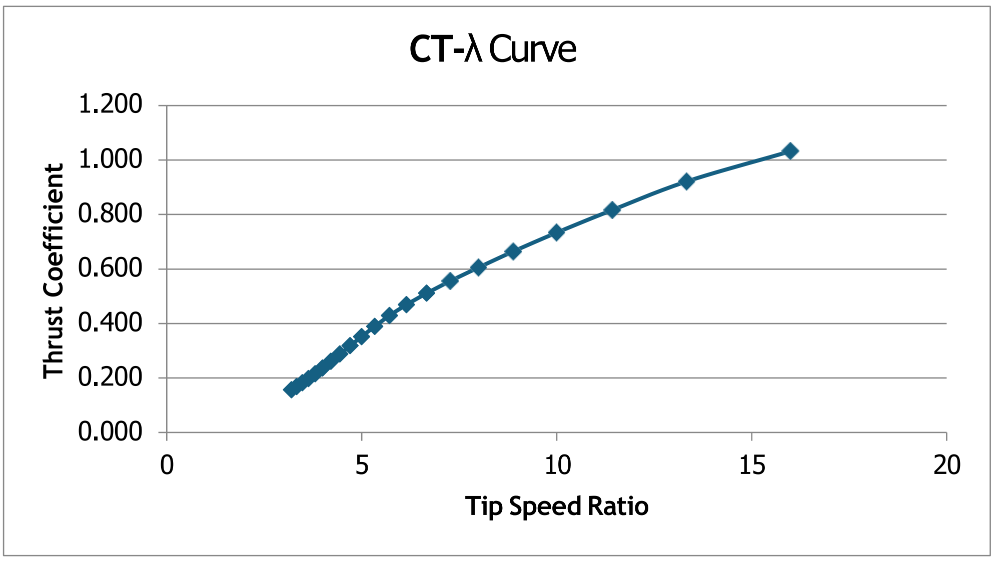

Wind Speed Vs Rotational Speed
Rotational Speed (rad/sec)
1.8
1.6
1.4
1.2
1
0.8
0.6
0.4
0.2
0
0
5
10
15
20
25
30
Wind Speed (m/sec)

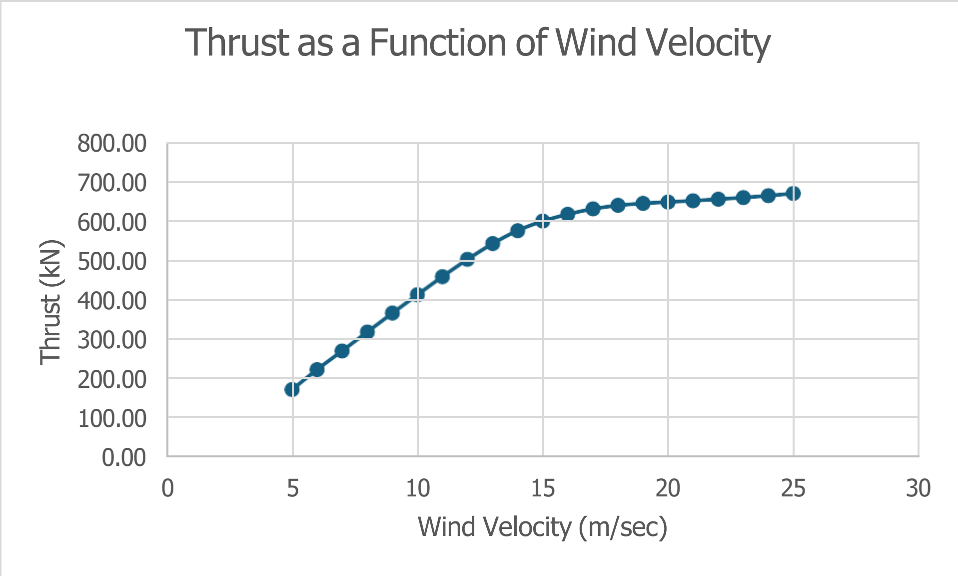

Pitch Angle
Pitch Angle (degree)
30
25
20
15
10
5
0
-5
0
5
10
15
20
25
30
Wind Speed (m/sec)

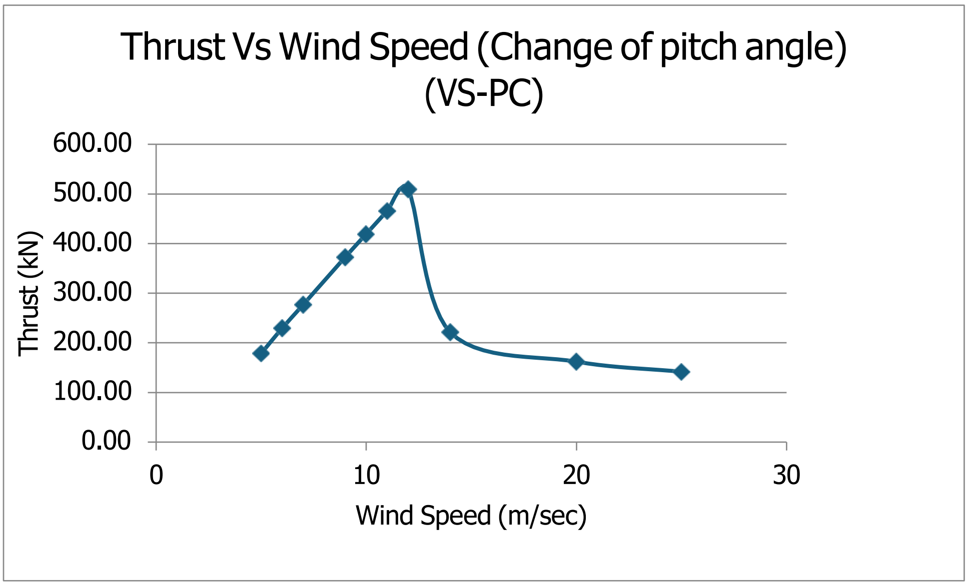
TSR nominal	Wind Speed	Thurst	Thurst Coeffi	Power	Force	Wind Power	Power Coefficient
9.60	5	178.70	1.032	284.09	173.0925	865.4625	0.328
8.00	6	229.64	0.921	490.92	249.2532	1495.5192	0.328
7.27	7	277.07	0.817	779.56	339.2613	2374.8291	0.328
7.27	8	325.08	0.734	1163.65	443.1168	3544.9344	0.328
7.27	9	372.71	0.665	1656.84	560.8197	5047.3773	0.328
7.27	10	419.46	0.606	2272.76	692.37	6923.7	0.328
7.27	11	465.77	0.556	2500.00	837.7677	9215.4447	0.271
6.67	12	509.46	0.511	2500.00	997.0128	11964.154	0.209
6.15	13	549.32	0.469	2500.00	1170.1053	15211.369	0.164
5.71	14	582.65	0.429	2500.00	1357.0452	18998.633	0.132
5.33	15	606.48	0.389	2500.00	1557.8325	23367.488	0.107
5.00	16	624.09	0.352	2500.00	1772.4672	28359.475	0.088
4.71	17	637.58	0.319	2500.00	2000.9493	34016.138	0.073
4.44	18	646.47	0.288	2500.00	2243.2788	40379.018	0.062
4.21	19	651.62	0.261	2500.00	2499.4557	47489.658	0.053
4.00	20	655.16	0.237	2500.00	2769.48	55389.6	0.045
3.81	21	658.43	0.216	2500.00	3053.3517	64120.386	0.039
3.64	22	662.27	0.198	2500.00	3351.0708	73723.558	0.034
3.48	23	666.50	0.182	2500.00	3662.6373	84240.658	0.030
3.33	24	671.21	0.168	2500.00	3988.0512	95713.229	0.026
3.20	25	676.98	0.156	2500.00	4327.3125	108182.81	0.023

Table for the wind speed, power, tip speed ratio, thrust, power coefficient and thrust coefficient



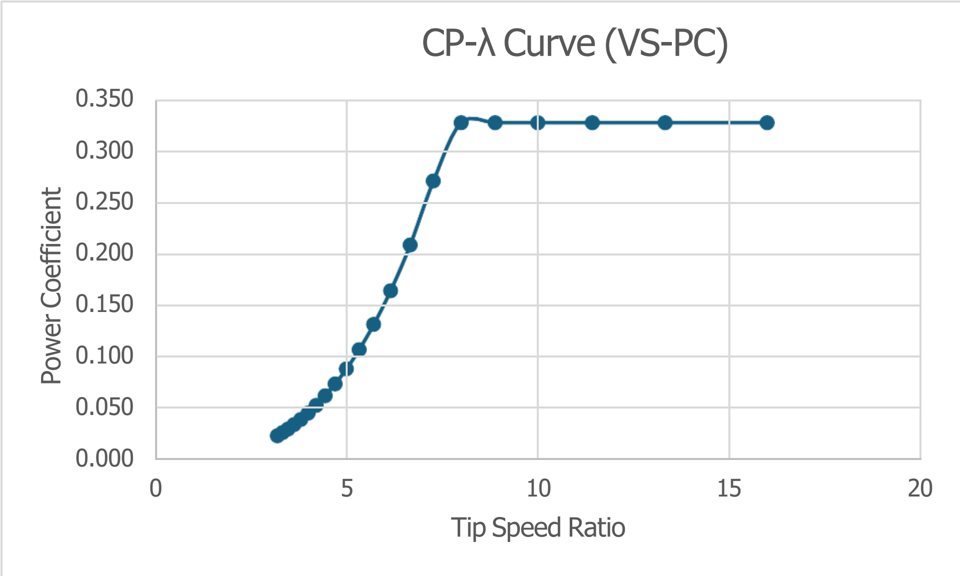


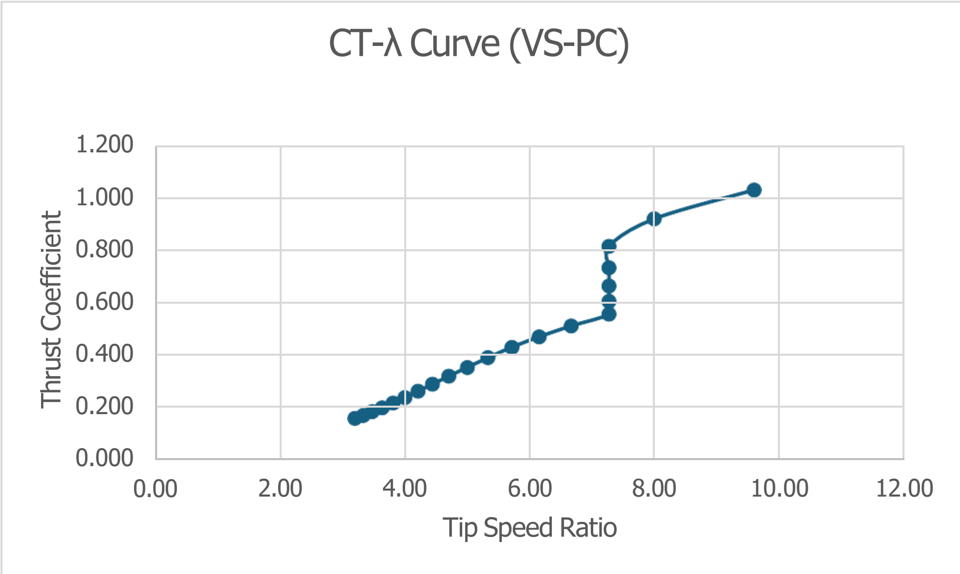




Wind Speed	Pitch Angle	Thrust
5	0	178.70
6	0	229.64
7	0	277.07
9	0	372.71
10	0	419.46
11	0	465.77
12	0	509.46
14	12.73	221.22127
20	21.25	162.04736
25	26.95	141.73189

Table for Wind speed and Thrust with a change in pitch angle

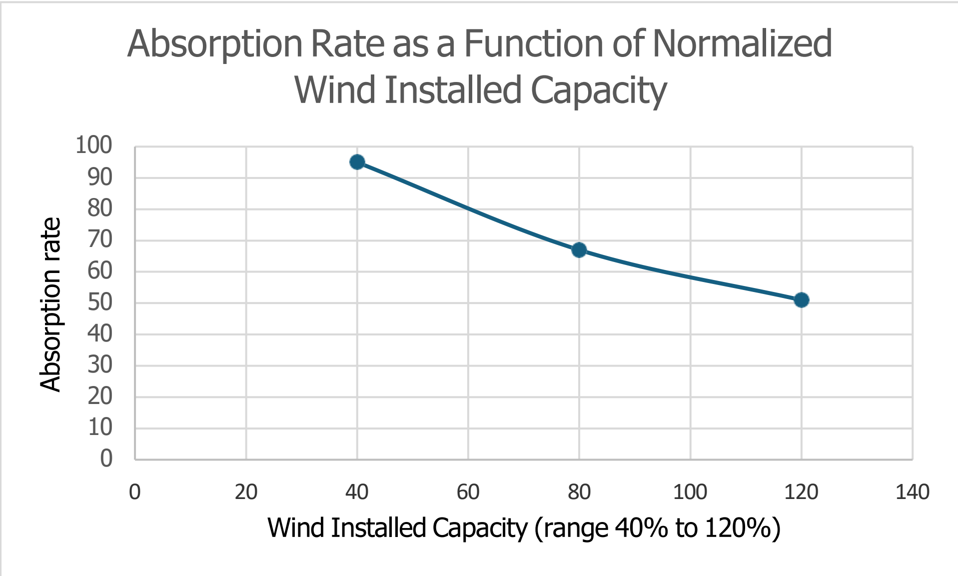


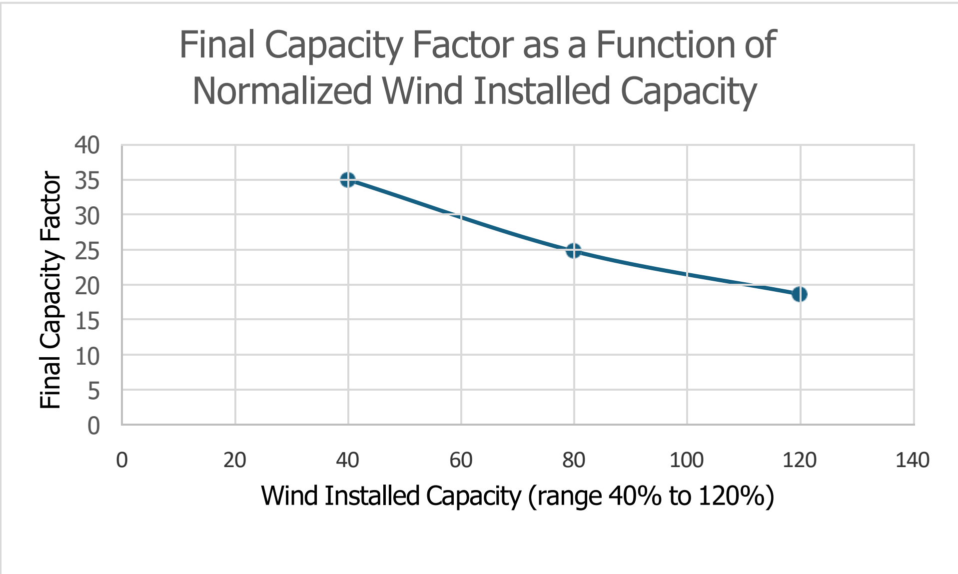


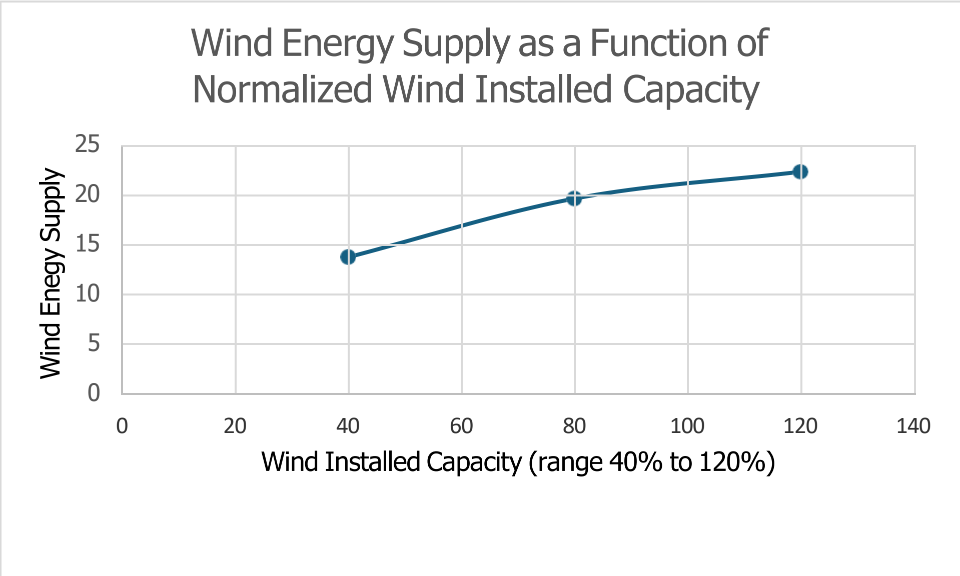
3. Wind Penetration Limits in Non-Interconnected Power System (i.e. island).

The number of wind turbines was assumed and the rated power of the wind turbine was added to the excel tool. The demand was been copied from the island that was given in the excel sheet. A probabilistic approach was applied and for three different cases for wind installed capacity 40%, 80%, 120%. The wind supply, capacity factor real and absorption rate using the wind curtailment excel tool. Graphs were plotted where the wind installed capacity was measured in the horizontal axis and the parameters which were being calculated (wind supply, capacity factor real and absorption rate) formed the vertical axis.

Wind Installed Capacity	40	80	120
Number of turbines	5	10	15
Wind Supply	13.8	19.7	22.4
Capacity Factor Real	35.36	25.17	19.04
Absorption Rate	95	67	51







4. Wind Capacity Credit.

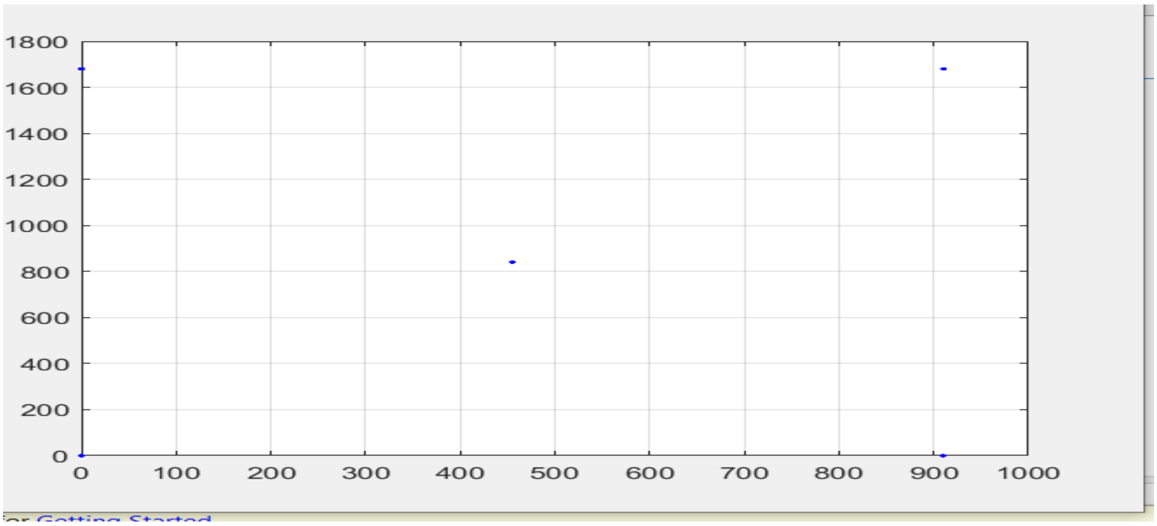
Wind Installed capacity was calculated as 80% of the annual mean load. The LOLP (in %) and the LOEE (in MWh) was calculated in two cases one before the wind energy installation in the autonomous power system and another is after the wind energy installation in the autonomous power system. The number of diesel units (15) was added to the excel tool of wind capacity credit and assumed that 90% of the diesel units are made available. The number of wind turbines were also adjusted and then two values of wind capacity was taken 80 % (15.3MWh) and 120% (23MWh) and the wind capacity credit was calculated from the tools as 24% and 16% respectively.

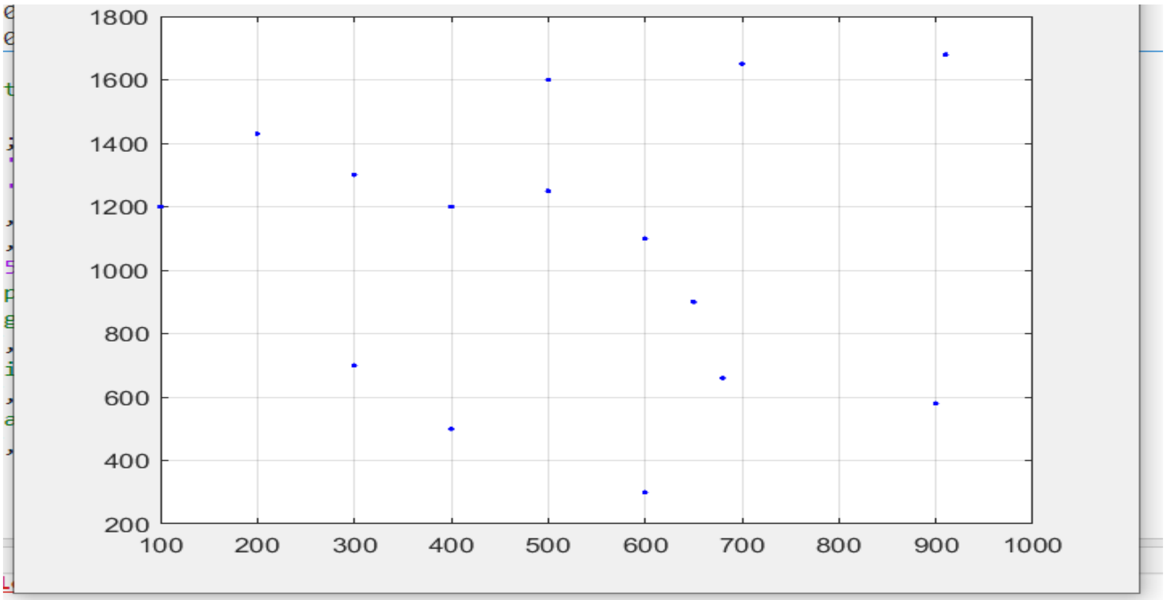
As percentage of the Average Load	80% (15.3MW)	120% (23MW)
ELCC	3.67	3.68
LOLE System after wind installation	0.061504%	0.0511%
Wind installed capacity (MWh)	15.3	23
Capacity Credit	24%	16%
LOEE (MWh)	16.70	14.67
ELCC (initial)	0	0

5. Wake Effect Losses- Wind Farm Layout Design.

L=910m, H=1680m

At first the number of wind turbines was assumed as 1. There were four notepad files which were required for finding the efficiency of the wake effect losses. At first the number of wind turbines, hub height and the rotor speed was adjusted in the dfile then the blade excel sheet was referred and the wind velocity with corresponding power and thrust coefficient was taken from that excel file and copied in the power_ct notepad file. The wind rose file contains the wind rose data from the first step. The task notepad should have the number of wind turbines and the plot of the wind turbine was created in this file. This steps were then repeated for the 5, 10 and 15 wind turbines and the result file that was created by running the wale effect application which contains the annual energy production, losses and the capacity factor. The wake effect loss was noted down zero for the one wind turbine but for the other cases there are wake effect losses. The percentage of wake loss should not exceed 5%. The calculations were based on two wake models Abramovich and GCL model. The efficiency due to wake effect loss was noted as 94% when tested using Abramovich model and 97% when using GCL model. A profile of wind speed was drawn at a distance of 3D to 5D with one wind turbine and two wind turbines located in a distance of 4D. In the following three diagrams the arrangements for 5,15 and 10 wind turbines are made in order to produce the minimum wake losses.

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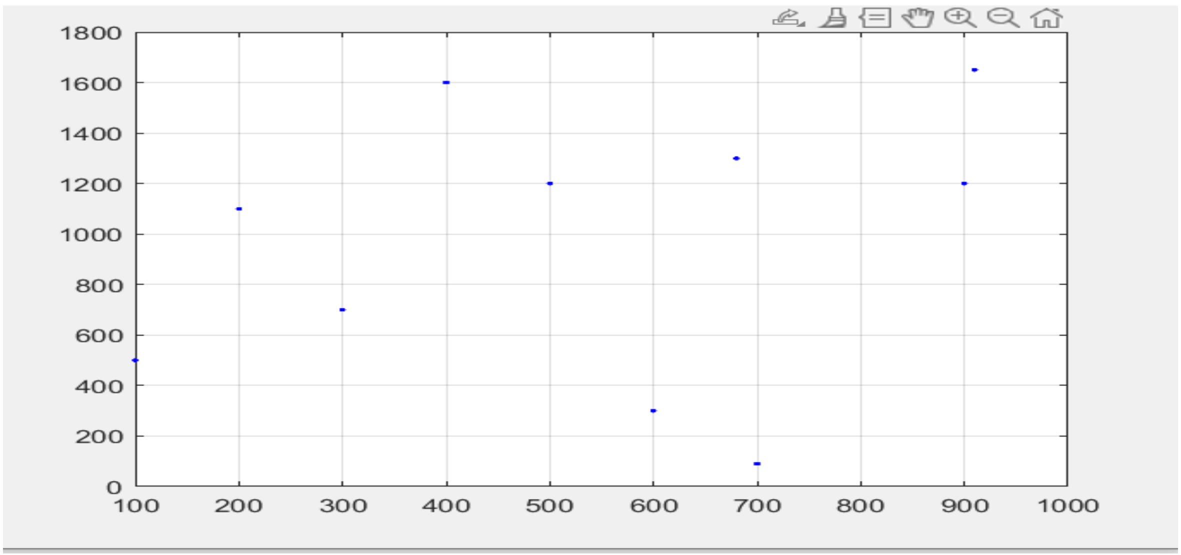
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900
1000

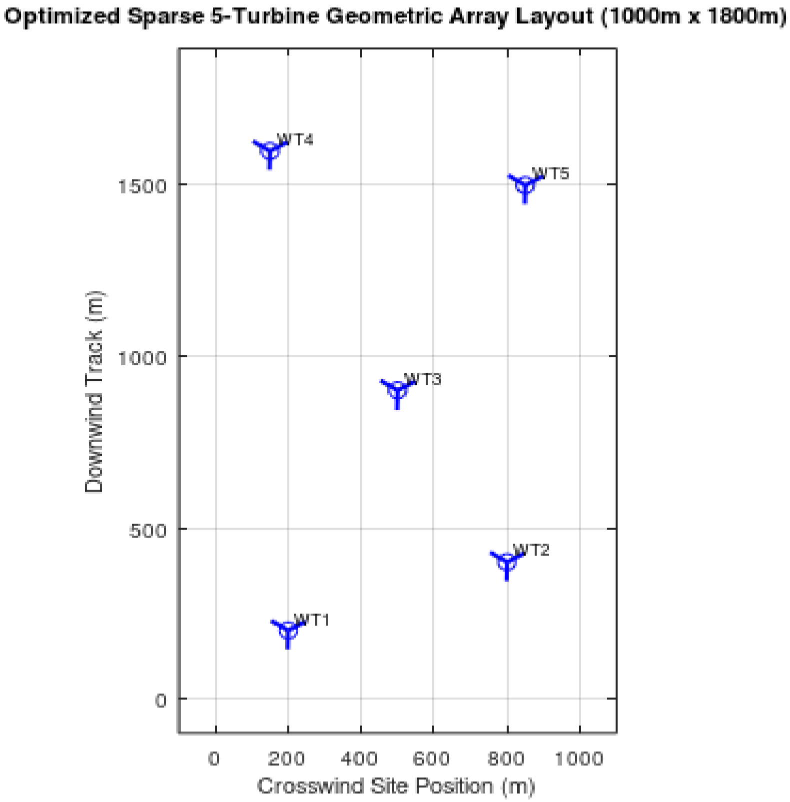
ABRAMOVICH MODEL

Number of wind turbines	1	5	10	15
Annual Energy Production	8355039.33	40004015.062	78774134.08	114158144.348
Losses	0.0000	1785715.597	4805327.232	11211047.628
Capacity Factor	0.382	0.365	0.36	0.348
Efficiency	-	0.96	0.94	0.90

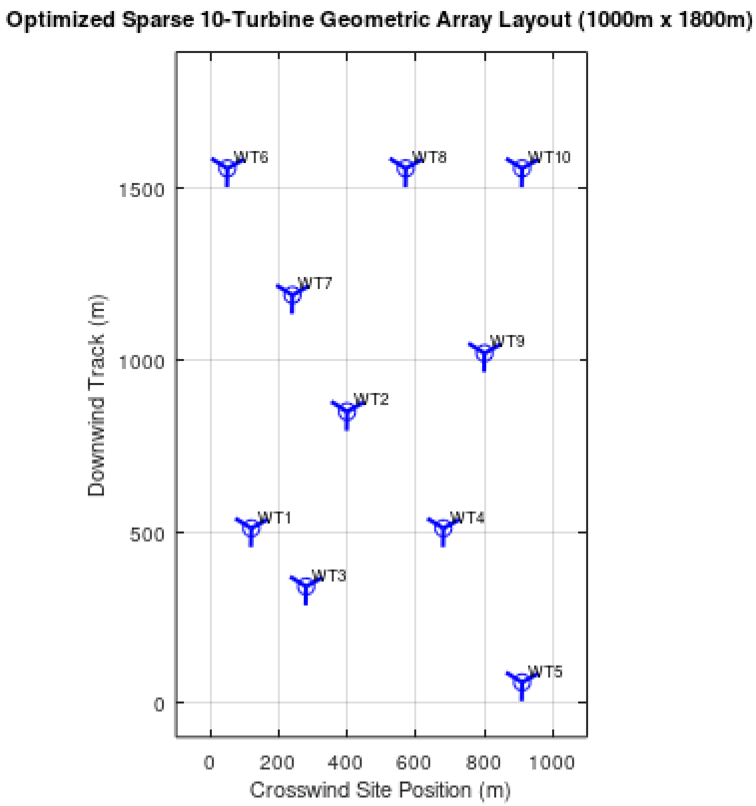
GCL MODEL

Number of wind turbines	1	5	10	15
Annual Energy Production	8357946.132	40622271.508	80196703.81	114770497.511
Losses	0.000	1167459.151	3382757.50	10598694.46
Capacity Factor	0.382	0.371	0.366	0.349
Efficiency	-	0.97	0.96	0.90

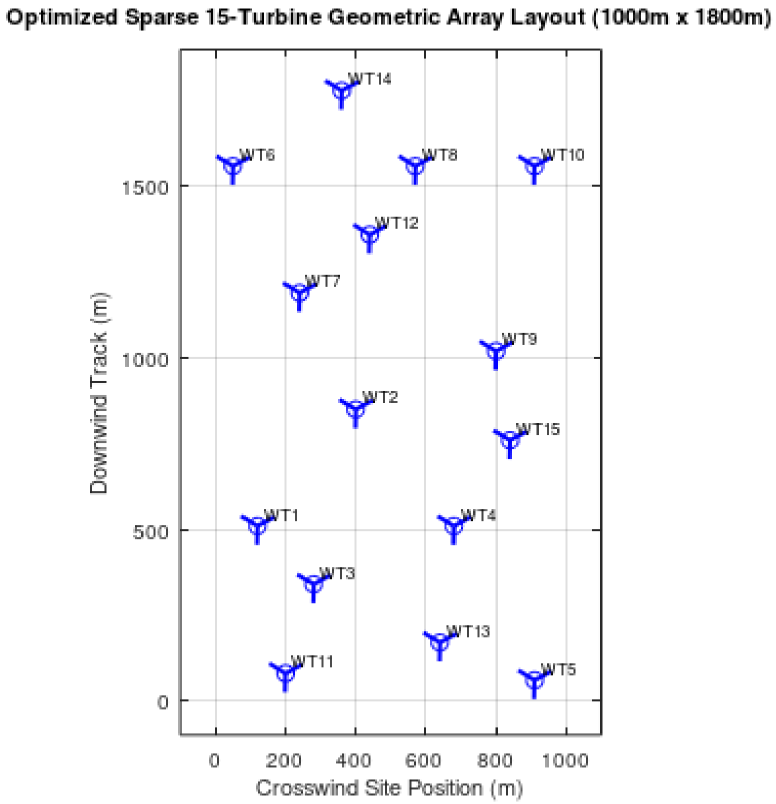
Position of 5 wind turbines



Position of 10 wind turbines



Position of 15 wind turbines



D=100m, a=0.05

Speed	Distance	R(x)	U(x)	U(x,r)	Ut	Power	Thrust	Slope
6m/sec	3D	183.8	1.79	4.36	3.11	-	0.73	0.446
12m/sec	5D	102	3.36	9.34	8.4	1719.06	0.51	0.104

6. Financial Evaluation.

Three wind installed capacity have been taken 40%, 80% and 120% of the maximum energy. The capacity factor was calculated in step 2 was inserted in the excel tool of financial evaluation. The absorption rate of the three cases (40%, 80%, 120% wind installed capacity) was inserted in the excel and also the efficiency due to wake effect loss was entered in the excel. The operational cost was calculated and the interest in each year was calculated. Finally the NPV and IRR were calculated for 10 years and 20 years. The payback time was calculated by performing the sum of the net annual cash flow until the sum is changing from negative to positive. As the number of turbine increases the IRR tends to become negative.

	Wind Farm Capacity	IRR (10 years)	NPV (thousand €) (10 years)	PBP (years)	IRR (20 years)	NPV (thousand €) (20 years)	LCoE (10 years)	LCoE (20 years)	No of WT
40%	12.5	0.15	1898.1	6	0.22	8787.0	0.07	0.05	5
80%	25	0.04	-3688.6	10	0.12	6264.2	0.11	0.08	10
120%	37.5	-0.10	-11994.3	15	0.08	2400.1	0.15	0.10	15

The wind farm that consists of 5 wind turbines will be a profitable investment as the IRR and NPV are positive and LCoE is less than the market price which is 0.08. The wind farm with 10 wind turbines will have marginal profitability as the LCoE is equal to the market price and IRR is almost zero for 10 years. Lastly the wind farm with 15 wind turbines is a non profitable investment as the IRR is negative, NPV is negative and the LCoE is higher than the market value.

Appendices:

M1-M240.xls

Blade 2023.xls

Cost Model2022.xls

Performance.xls

Wind curtailment project.xls

Capacity credit project.xls

Wind financial evaluation.xls

N1-N240.xls

Weibull energy calculation.xls

Raft.exe

Wake effect.exe